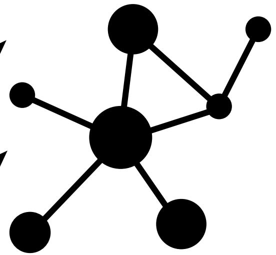
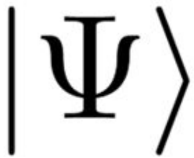
Toy data

PPI data

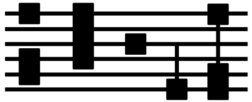


Sparse network

Embedding

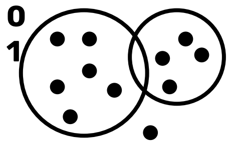


QMME
(amplitude encoding of neighbors statistics)

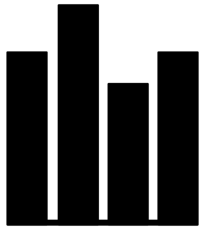


Embedding state

Classification



QC
(distance-based non-parametric classification)



Measurements

Repeat for all unpredicted nodes, or in parallel

Pre-Assumptions:
- Sparse network
- Distance as criteria

Pre-Requirements:
Access to data
- Oracle L - Features
- Oracle K - Degrees
- Oracle L - Adjacency matrix
- Oracle Y - Labels (training set)

Key papers:
- Prediction of cancer driver genes through network-based moment propagation of mutation scores
- MM-GNN: Mix-Moment Graph Neural Network towards Modeling Neighborhood Feature Distribution
- Quantum classifier with tailored quantum kernel

Key areas:
- Quantum computing
    - Sparse networks
    - Amplitude encoding mapping
    - Quantum classifiers
    - Complexity theory
- Graph theory/complex networks
- PPI networks/gene expression
- Statistics/ML/data science

Key goals:
- Theory
    - Full pipeline (embedding + classification)
    - Non-iterative/non-variational
- Validation
    - Toy verification + real potential

Mathematical + proof of concept
Won’t cure cancer, proof-method

Key challenges:
- Oracle assumptions
    - Regular graph/graph augmentation
    - Angle for rotation gates
    - Access to QRAM (?)
- Classifier too simple
- Might be more costly (complexity-wise)